AUDIT REPORT

fleet-routing-cvrp: demo video audit report

A slide-by-slide record of every number, assumption and calculation in the demo video, with its source.

Item	Value
Repository	https://github.com/kavya2693/fleet-routing-cvrp
Numbers taken from commit	`c9b3c09d094cd863c61c2af426513b6d9b33ba68` (outputs last changed in `c9b3c09 (2026-10-13 21:56:42 +0400)`)
Video	`docs/demo/demo.mp4` · 84.0 s · 1920x1080 · 30 fps · silent
Generated	2026-10-14 13:44
Automated checks	52 pass · 0 fail · 4 manual (of 56 figures)
Assumptions	16 (A-01 to A-16)

How to use this report

Every number or claim on screen has an ID (F-01, F-02, ...) and every modelling assumption has an ID (A-01, ...). Cite the ID. Each figure lists its source file, the exact column, row or key, the formula for derived numbers, and an automated check result. The change-impact matrix (section 5) shows which slides and files change if an input or assumption changes.

Types: direct = read straight from a repo output; derived = computed from direct values (formula given); illustrative = drawing or example, not data; general knowledge = context not produced by the repo.

Regenerate

```
make run                                    # outputs/full/*.csv, results.json, route_map.png (~18 min)
python3 docs/demo/source/tools/darkmap.py outputs/full/route_map.png docs/demo/source/assets
PYTHONPATH=src .venv/bin/python docs/demo/source/tools/export_dispatch_sheet.py
docs/demo/source/assets/dispatch_sheet.json
python3 docs/demo/source/tools/inject_sheet.py docs/demo/source/assets/dispatch_sheet.json
docs/demo/source/compositions/value.html --rows 12,24,30,34,38,53 --animate 24,30,34,53
cd docs/demo/source && npx --yes hyperframes@0.8.156 check && npx --yes hyperframes@0.8.156 render -f 30 -q
looks -o renders/fleet-routing-cvrp-master.mp4 && cd -
ffmpeg -y -i docs/demo/source/renders/fleet-routing-cvrp-master.mp4 -c:v libx264 -preset slow -crf 25
-pix_fmt yuv420p -movflags +faststart -an docs/demo/demo.mp4
python3 docs/demo/source/tools/build_audit_report.py docs/demo/source/audit/facts.yaml   # stills, checks,
MD/DOCX/PDF
```

1. Figures

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-01	3 industry chips	01	author's framing	-	-	general knowledge	manual: Context statement, not a repo output.
F-02	4 problem cards	01	README 'The problem' (problems 2-4)	README lines 16-27	-	general knowledge	✓ verified
F-03	300 trucks	02, 05, 08	outputs/full/results.json	trucks	-	direct	✓ verified
F-04	4 depots	02, 08	outputs/full/results.json	depots	-	direct	✓ verified
F-05	Up to 2,800 orders a day	02	data/full/dispatch_orders.csv	orders per date (max = 2,785 on 2025-12-08)	max daily order count = 2,785, rounded up to the nearest hundred = 2,800	derived	✓ verified
F-06	every stop has a delivery window	02	data/full/dispatch_orders.csv	tw_start, tw_end (no missing values)	count of missing windows = 0	direct	✓ verified
F-07	trailer capacity + shift limits	02	src/fleet_routing/routing.py:202, :205	capacity + time dimensions	-	direct	✓ verified
F-08	>5% of peak shipments misrouted	02	outputs/full/risk_scores.csv; README line 26-27	model=xgboost, period=peak, base_rate (0.0531)	base_rate x 100 = 5.3% > 5%	direct	✓ verified
F-09	OR-Tools, 6 s per depot / solve budget 6.0 s	03, 05	outputs/full/results.json	solver_seconds_per_depot_day	-	direct	✓ verified
F-10	EVERY MONDAY	03	README line 43	README	-	direct	✓ verified
F-11	windows kept / fewer km / fewer trucks	03	outputs/full/routing_summary.csv	arm in {sweep, ortools_risk}, window=peak: on_time_pct, total_km, vehicle_days	on_time_pct = 100; total_km(ortools_risk) < total_km(sweep); vehicle_days(ortools_risk) < vehicle_days(sweep)	derived	✓ verified
F-12	build, score, improve	03	src/fleet_routing/routing.py:229-230	PATH_CHEAPEST_ARC + GUIDED_LOCAL_SEARCH	-	direct	✓ verified
F-13	Later stops are riskier	04	src/fleet_routing/data.py:238-241; outputs/full/results.json position_effect_logodds_per_stop	0.09 generated, 0.094 estimated	estimated slope > 0	direct	✓ verified
F-14	-4.2% misroutes	04	outputs/full/routing_summary.csv	window=peak, misroute_pct: ortools_risk 4.3871, ortools 4.5810	(4.3871 - 4.5810) / 4.5810 x 100 = -4.23%	derived	✓ verified
F-15	+4.0% km	04	outputs/full/routing_summary.csv	window=peak, total_km: ortools_risk 113,937.7, ortools 109,607.3	(113,937.7 - 109,607.3) / 109,607.3 x 100 = +3.95%	derived	✓ verified
F-16	2 yrs x 40 zones	05, 08	data/full/zone_demand.csv; outputs/full/results.json zones	731 daily rows 2024-01-01..2025-12-31; zones = 40	-	direct	✓ verified
F-17	200k history	05	data/full/shipments_history.csv	row count	-	direct	✓ verified
F-18	24k	05	outputs/full/results.json	dispatch_orders = 24,394	24,394 rounded to thousands	direct	✓ verified
F-19	+0.094 log-odds / stop	05	outputs/full/results.json	position_effect_logodds_per_stop	-	direct	✓ verified
F-20	forecast.py, risk.py, simulate.py, routing.py, replay(), data.py	05	src/fleet_routing/	files + def replay	-	direct	✓ verified
F-21	call top 5% before loading	05, 09	src/fleet_routing/config.py:93	ADDRESS_CHECK_SHARE = 0.05	-	direct	✓ verified
F-22	Depot 3 · dispatch 2025-12-09	06	data/full/depots.csv (busiest depot); src/fleet_routing/cli.py:121-125; config peak window	depot with most vehicles = 3; first peak dispatch = 2025-12-08 + 1 day	-	direct	✓ verified
F-23	104 trucks	06	outputs/full/route_map.png (left panel title)	plot title	-	direct	manual: Read from the PNG title; the repo does not save per-depot-day routes. Cross-check: depot 3 fleet = 104 (depots.csv).
F-24	11,368 km	06	outputs/full/route_map.png (left panel title)	plot title	-	direct	manual: Read from the PNG title; not persisted in any CSV.
F-25	93 trucks	06	outputs/full/route_map.png (right panel title); re-run in demo assets/dispatch_sheet.json	trucks_used	-	direct	✓ verified
F-26	8,574 km	06	outputs/full/route_map.png (right panel title)	plot title	-	direct	manual: Read from the PNG title. The single-day re-run

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
							(assets/dispatch_sheet.json) gives 8,412.7 km because the solver stops on wall-clock time (A-04).
F-27	-24.6% km	06	F-24, F-26 (plot titles)	8,574 vs 11,368	(8,574 - 11,368) / 11,368 x 100 = -24.58%	derived	✓ verified
F-28	13,554 orders	06	outputs/full/routing_summary.csv	arm=sweep, window=peak, orders	-	direct	✓ verified
F-29	140,683 km	06, 08	outputs/full/routing_summary.csv	arm=sweep, window=peak, total_km	-	direct	✓ verified
F-30	109,607 km	06, 08	outputs/full/routing_summary.csv	arm=ortools, window=peak, total_km	-	direct	✓ verified
F-31	-22.1%	06, 08	outputs/full/routing_summary.csv	F-30 vs F-29	(109,607.3 - 140,683.4) / 140,683.4 x 100 = -22.09%	derived	✓ verified
F-32	93 trucks	07	demo assets/dispatch_sheet.json (re-run of the repo's routing for Depot 3, 2025-12-09)	trucks_used (rostered 104)	-	direct	✓ verified
F-33	910 orders	07	demo assets/dispatch_sheet.json; data/full/dispatch_orders.csv	orders on 2025-12-08 for depot 3	-	direct	✓ verified
F-34	all inside their delivery windows	07	demo assets/dispatch_sheet.json	on_time = delivered = orders	910 / 910	direct	✓ verified
F-35	8,413 km	07	demo assets/dispatch_sheet.json	total_km = 8,412.7	rounded	direct	✓ verified
F-36	T-013, T-025, T-031, T-035, T-039, T-054	07	demo assets/dispatch_sheet.json, injected into compositions/value.html	vehicles 12, 24, 30, 34, 38, 53	truck id = vehicle index + 1; depart = SHIFT_START	direct	✓ verified
F-37	+ 87 more trucks	07	F-32	93 - 6 rows	93 - 6 = 87	derived	✓ verified
F-38	-8.1%	08	outputs/full/routing_summary.csv	vehicle_days: ortools 1,339 vs sweep 1,457	(1,339 - 1,457) / 1,457 x 100 = -8.10%	derived	✓ verified
F-39	1,457	08	outputs/full/routing_summary.csv	arm=sweep, window=peak, vehicle_days	-	direct	✓ verified
F-40	1,339	08	outputs/full/routing_summary.csv	arm=ortools, window=peak, vehicle_days	-	direct	✓ verified
F-41	0.208	08	outputs/full/forecast_scores.csv	model=xgboost, period=all, wape	-	direct	✓ verified
F-42	ETS 0.260	08	outputs/full/forecast_scores.csv	model=ets, period=all, wape	-	direct	✓ verified
F-43	naive 0.344	08	outputs/full/forecast_scores.csv	model=seasonal_naive, period=all, wape	-	direct	✓ verified
F-44	8.7%	08	outputs/full/roster_scores.csv	model=xgboost, period=peak, short_depot_days_pct	-	direct	✓ verified
F-45	ETS forecast: 53.8%	08	outputs/full/roster_scores.csv	model=ets, period=peak, short_depot_days_pct	-	direct	✓ verified
F-46	16.6%	09	outputs/full/address_checks.csv	ranked_by=rule, window=peak, reduction_pct	-	direct	✓ verified
F-47	16.4%	09	outputs/full/address_checks.csv	ranked_by=xgboost, window=peak, reduction_pct	-	direct	✓ verified
F-48	15.8%	09	outputs/full/address_checks.csv	ranked_by=logistic, window=peak, reduction_pct	-	direct	✓ verified
F-49	1.1%	09	outputs/full/address_checks.csv	ranked_by=random, window=peak, reduction_pct	-	direct	✓ verified
F-50	ROC-AUC 0.645	09	outputs/full/risk_scores.csv	model=xgboost, period=all, roc_auc	-	direct	✓ verified
F-51	ROC-AUC 0.647	09	outputs/full/risk_scores.csv	model=logistic, period=all, roc_auc	-	direct	✓ verified
F-52	ceiling AUC 0.647	09	outputs/full/risk_scores.csv	model=ceiling, period=all, roc_auc	-	direct	✓ verified
F-53	MIT	10	LICENSE line 1	-	-	direct	✓ verified
F-54	~18 min	10	outputs/full/results.json	runtime_seconds = 1063	1063 / 60 = 17.7 min	derived	✓ verified
F-55	OR-Tools · XGBoost · statsmodels · Python	10	pyproject.toml dependencies	ortools, xgboost, statsmodels	-	direct	✓ verified
F-56	$ make run	10	Makefile	run target	-	direct	✓ verified

2. Automated check run (2026-10-14 13:44)

ID	Expected on slide	Recomputed from repo	Result
F-01	3 industry chips	-	manual
F-02	true	true	✓ pass
F-03	300	300	✓ pass
F-04	4	4	✓ pass
F-05	true	true	✓ pass
F-06	true	true	✓ pass
F-07	true	true	✓ pass
F-08	true	true	✓ pass
F-09	6.0	6.0	✓ pass
F-10	true	true	✓ pass
F-11	true	true	✓ pass
F-12	true	true	✓ pass
F-13	true	true	✓ pass
F-14	-4.2%	-4.2%	✓ pass
F-15	+4.0%	+4.0%	✓ pass
F-16	true	true	✓ pass
F-17	200,000	200,000	✓ pass
F-18	24k	24k	✓ pass
F-19	0.094	0.094	✓ pass
F-20	true	true	✓ pass
F-21	true	true	✓ pass
F-22	true	true	✓ pass
F-23	104 trucks	-	manual
F-24	11,368 km	-	manual
F-25	93	93	✓ pass
F-26	8,574 km	-	manual
F-27	-24.6%	-24.6%	✓ pass
F-28	13,554	13,554	✓ pass
F-29	140,683	140,683	✓ pass
F-30	109,607	109,607	✓ pass
F-31	-22.1%	-22.1%	✓ pass
F-32	93	93	✓ pass
F-33	true	true	✓ pass
F-34	true	true	✓ pass
F-35	8,413	8,413	✓ pass
F-36	true	true	✓ pass
F-37	87	87	✓ pass
F-38	-8.1%	-8.1%	✓ pass
F-39	1,457	1,457	✓ pass
F-40	1,339	1,339	✓ pass
F-41	0.208	0.208	✓ pass
F-42	0.260	0.260	✓ pass
F-43	0.344	0.344	✓ pass
F-44	8.7%	8.7%	✓ pass
F-45	53.8%	53.8%	✓ pass
F-46	16.6%	16.6%	✓ pass
F-47	16.4%	16.4%	✓ pass
F-48	15.8%	15.8%	✓ pass
F-49	1.1%	1.1%	✓ pass
F-50	0.645	0.645	✓ pass
F-51	0.647	0.647	✓ pass
F-52	0.647	0.647	✓ pass
F-53	true	true	✓ pass
F-54	~18 min	~18 min	✓ pass
F-55	true	true	✓ pass
F-56	true	true	✓ pass

3. Assumptions register

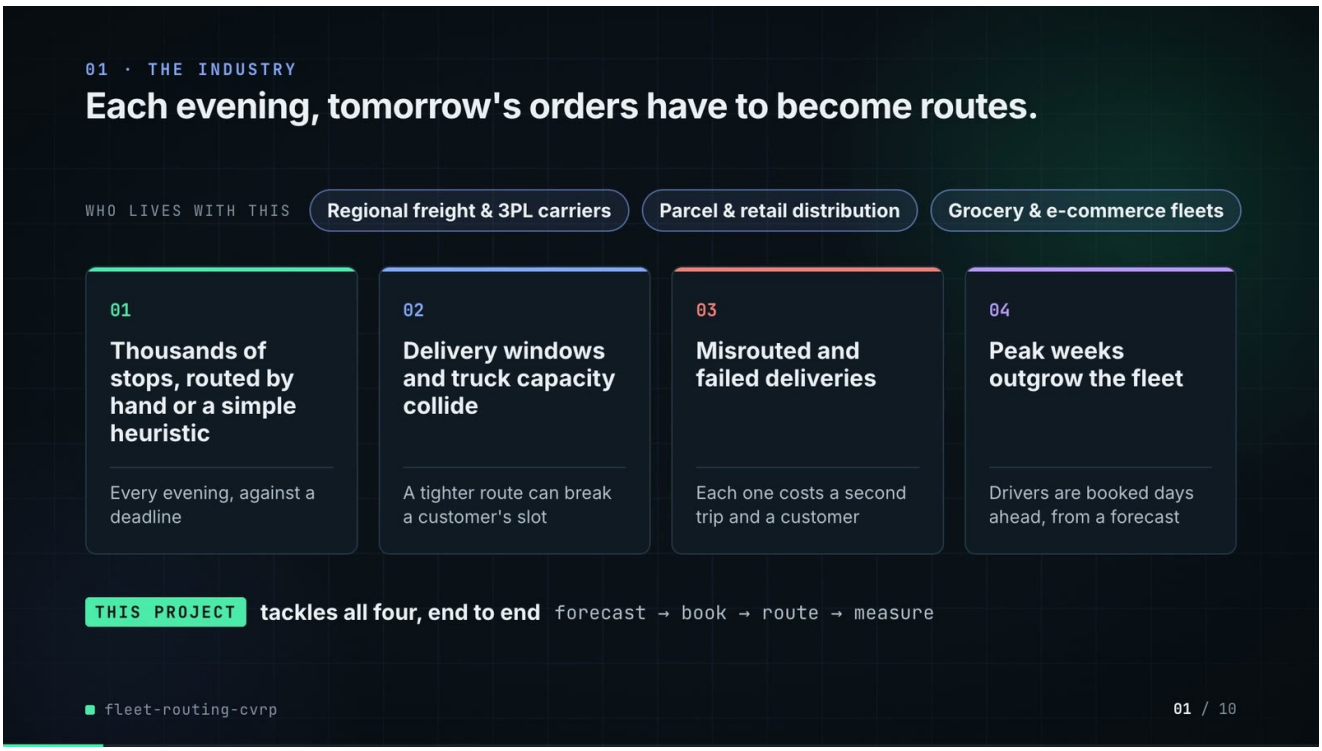
ID	Assumption	Source	Figures affected	Slides affected
A-01	All data is synthetic, produced by a seeded generator sized to the production problem (300 trucks, 4 depots, 40 zones, 2 years of demand, 200,000 historical shipments). The production data is proprietary.	src/fleet_routing/config.py:32-46 (FULL profile, seed=7); scripts/generate_data.py; README 'The data'	all figures	slides 01, 02, 03, 04, 05, 06, 07, 08, 09, 10
A-02	Misroute risk rises with route position: the generator adds 0.09 log-odds per stop past the first, capped at 15 stops. The pipeline re-estimates it from history (0.094).	src/fleet_routing/data.py:238-241 (position_logit); README 'Known limitations' (biggest gap)	F-13, F-14, F-15, F-19, F-46, F-47, F-48, F-49	slides 04, 05, 09
A-03	A misroute is priced at 150 km inside the risk-priced router (cost per stop = 150 km x 0.094 x p(1-p) per	src/fleet_routing/config.py:90 (MISROUTE_COST_KM = 150.0); README 'The approach'	F-14, F-15, F-23, F-24, F-25, F-26, F-27, F-32, F-33, F-34,	slides 04, 06, 07

ID	Assumption	Source	Figures affected	Slides affected
	position past the first).		F-35, F-36	
A-04	Each OR-Tools solve is capped at 6 s wall-clock per depot-day (1.5 s on the small profile). Results vary slightly run to run and with machine load.	src/fleet_routing/config.py:45 (solver_seconds=6.0); routing.py:231; README 'Known limitations'	F-09, F-25, F-26, F-30, F-31, F-35, F-38, F-40	slides 03, 05, 06, 07, 08
A-05	Distances are straight lines x 1.3 (circuity) at a flat 50 km/h. No road network, traffic or driver breaks.	src/fleet_routing/config.py:74 (SPEED_KMH = 50.0), :76 (CIRCUITY = 1.3); README 'Known limitations'	F-24, F-26, F-27, F-29, F-30, F-31, F-35, F-36	slides 06, 07, 08
A-06	Drivers leave the depot from 06:00 and must be back by 18:00.	src/fleet_routing/config.py:72-73 (SHIFT_START, SHIFT_END); routing.py:205	F-07, F-36	slides 02, 07
A-07	Delivery windows are hard constraints in OR-Tools (it drops a stop rather than arrive late) and are ignored by the sweep baseline, so the on-time comparison is lopsided by design.	src/fleet_routing/ routing.py:205-225; README 'Known limitations'	F-11, F-34	slides 03, 07
A-08	A pre-dispatch call fixes the postcode, address and handwritten-label faults completely; it cannot fix a between-depots zone or a cross-dock relabel.	src/fleet_routing/risk.py:134-142 (address_check); README 'Known limitations'	F-46, F-47, F-48, F-49	slides 09
A-09	The dispatch office calls the top 5% of each day's consignees.	src/fleet_routing/config.py:93 (ADDRESS_CHECK_SHARE = 0.05)	F-21, F-46, F-47, F-48, F-49	slides 05, 09
A-10	Trucks booked per depot-day = ceil(forecast orders x 2.1 pallets/stop x 1.10 headroom / (mean capacity x 0.85 fill)), capped at the depot fleet.	src/fleet_routing/ simulate.py:36-45 (roster_size); config.py:82-84	F-44, F-45, F-32	slides 07, 08
A-11	Fleet mix: 70% tractors with 26 pallets, 30% rigids with 14 pallets; depot fleets 63 / 86 / 47 / 104.	src/fleet_routing/config.py:67-69; data/full/depots.csv	F-03, F-23, F-25, F-32, F-36	slides 02, 05, 06, 07, 08
A-12	Evaluation windows: quiet week 2025-10-13..17 and peak week 2025-12-08..12 (orders Mon-Fri, dispatched the next day, leftovers carried, up to 3 clear-out days).	src/fleet_routing/config.py:43, :95 (CLEAR_OUT_DAYS = 3)	F-28, F-28, F-29, F-30, F-31, F-38, F-39, F-40	slides 06, 08
A-13	Forecasts are backtested from 12 weekly (Monday) origins over the last 12 weeks of 2025; every feature is at least a week old.	src/fleet_routing/config.py:40, :44; forecast.py:166-183	F-10, F-41, F-42, F-43, F-44, F-45	slides 03, 08
A-14	Router cost terms: a dropped stop costs 1,000 km x (1 + days waited); each truck used costs 40 km.	src/fleet_routing/routing.py:32 (DROP_PENALTY_KM), :34 (VEHICLE_FIXED_KM)	F-25, F-26, F-30, F-35, F-40	slides 06, 07, 08
A-15	Video rounding: percentages to 1 decimal, km and counts to whole numbers with thousands separators, WAPE and AUC to 3 decimals; '24k', '200k', '~18 min' are rounded labels.	this audit (display convention)	F-14, F-15, F-18, F-27, F-31, F-38, F-41, F-42, F-43, F-44, F-45, F-46, F-47, F-48, F-49, F-50, F-51, F-52, F-54	slides 04, 05, 06, 08, 09, 10
A-16	Slide 01 (industry, everyday problems) is general logistics knowledge used as context. It is not a result of this repo and carries no numbers.	author's framing; README 'The problem' supports problems 2-4	F-01, F-02	slides 01

4. Slide by slide

Slide 01: The industry

0.0-7.2 s · composition `compositions/s01-context.html`



On-screen text (verbatim)

01 · THE INDUSTRY
Each evening, tomorrow's orders have to become routes.
WHO LIVES WITH THIS · Regional freight & 3PL carriers · Parcel & retail distribution · Grocery & e-commerce fleets
01 Thousands of stops, routed by hand or a simple heuristic — Every evening, against a deadline
02 Delivery windows and truck capacity collide — A tighter route can break a customer's slot
03 Misrouted and failed deliveries — Each one costs a second trip and a customer
04 Peak weeks outgrow the fleet — Drivers are booked days ahead, from a forecast
THIS PROJECT tackles all four, end to end · forecast → book → route → measure

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-01	3 industry chips	`author's framing` · -	-	manual
F-02	4 problem cards	`README 'The problem' (problems 2-4)` · README lines 16-27	-	✓

Assumptions: A-16

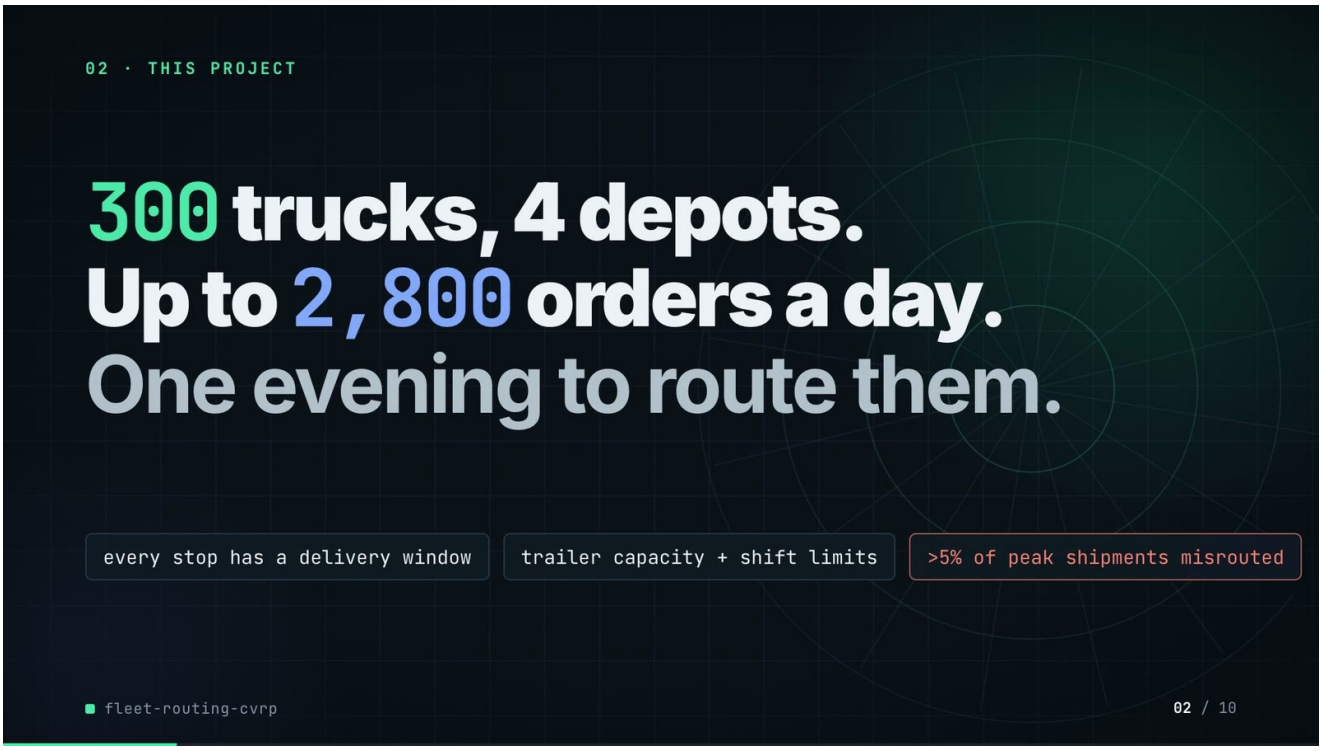
Illustrative, not data:

- The whole slide is context; no repo figures.

To change this slide: Edit the chips (#ctx-c1..3) and cards (#ctx-k1..4) in compositions/s01-context.html.

Slide 02: This project

6.8-11.6 s · composition `compositions/s02-hook.html`



On-screen text (verbatim)

02 · THIS PROJECT

300 trucks, 4 depots. Up to 2,800 orders a day. One evening to route them.

every stop has a delivery window · trailer capacity + shift limits · >5% of peak shipments misrouted

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	300 trucks	`outputs/full/results.json` · trucks	-	✓
F-04	4 depots	`outputs/full/results.json` · depots	-	✓
F-05	Up to 2,800 orders a day	`data/full/dispatch_orders.csv` · orders per date (max = 2,785 on 2025-12-08)	max daily order count = 2,785, rounded up to the nearest hundred = 2,800	✓
F-06	every stop has a delivery window	`data/full/dispatch_orders.csv` · tw_start, tw_end (no missing values)	count of missing windows = 0	✓
F-07	trailer capacity + shift limits	`src/fleet_routing/routing.py:202, :205` · capacity + time dimensions	-	✓
F-08	>5% of peak shipments misrouted	`outputs/full/risk_scores.csv; README line 26-27` · model=xgboost, period=peak, base_rate (0.0531)	base_rate x 100 = 5.3% > 5%	✓

F-05 note: Peak-week days have 2,622-2,785 orders. The README describes the peak as 'closer to 2,700' a day (an average); the slide states the upper bound.

Assumptions: A-01, A-06, A-11

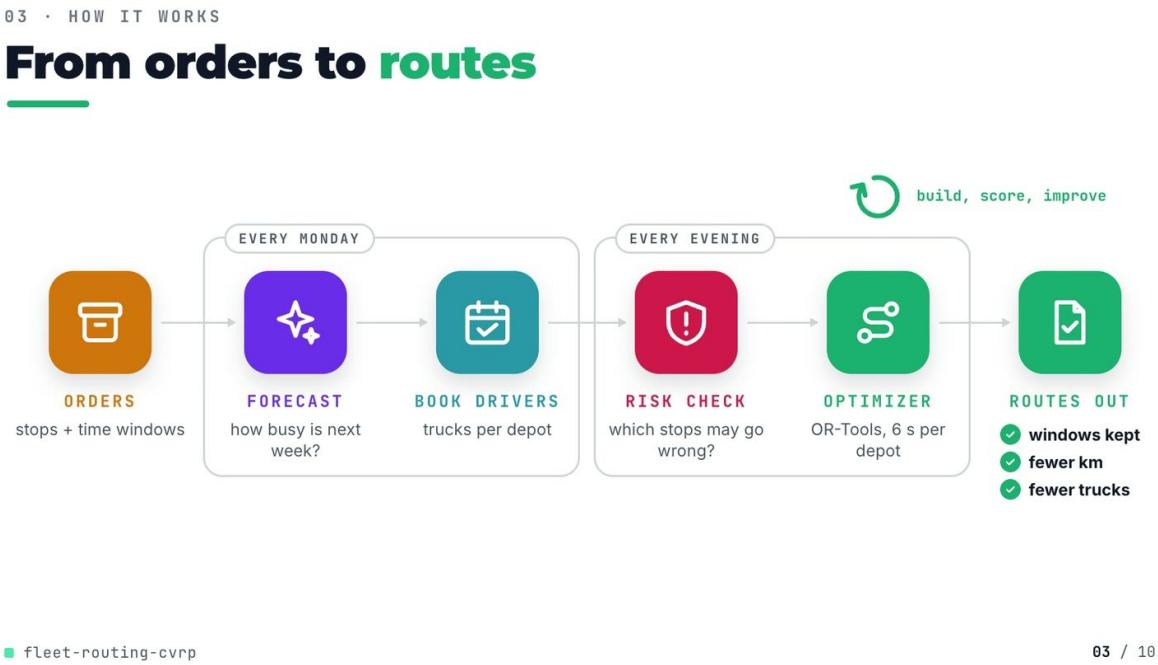
Illustrative, not data:

- Depot rings in the background are decoration.

To change this slide: Headline spans #hook-l1..l3 and chips #hook-f1..f3 in compositions/s02-hook.html; fleet/depots come from config.py FULL profile.

Slide 03: How it works (white chapter, slide A)

11.3-19.3 s · composition `compositions/explainer.html (CONFIG.blocks, CONFIG.groups, CONFIG.loop)`



On-screen text (verbatim)

03 · HOW IT WORKS

From orders to routes

ORDERS stops + time windows → [EVERY MONDAY: FORECAST how busy is next week? → BOOK DRIVERS trucks per depot] → [EVERY EVENING: RISK CHECK which stops may go wrong? → OPTIMIZER OR-Tools, 6 s per depot] → ROUTES OUT ✓ windows kept ✓ fewer km ✓ fewer trucks

build, score, improve (loop over the optimizer)

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-09	OR-Tools, 6 s per depot / solve budget 6.0 s	`outputs/full/results.json` · solver_seconds_per_depot_day	-	✓
F-10	EVERY MONDAY	`README line 43` · README	-	✓
F-11	windows kept / fewer km / fewer trucks	`outputs/full/routing_summary.csv` · arm in {sweep, ortools_risk}, window=peak: on_time_pct, total_km, vehicle_days	on_time_pct = 100; total_km(ortools_risk) < total_km(sweep); vehicle_days(ortools_risk) < vehicle_days(sweep)	✓
F-12	build, score, improve	`src/fleet_routing/routing.py:229-230` · PATH_CHEAPEST_ARC + GUIDED_LOCAL_SEARCH	-	✓

Assumptions: A-04, A-07, A-13

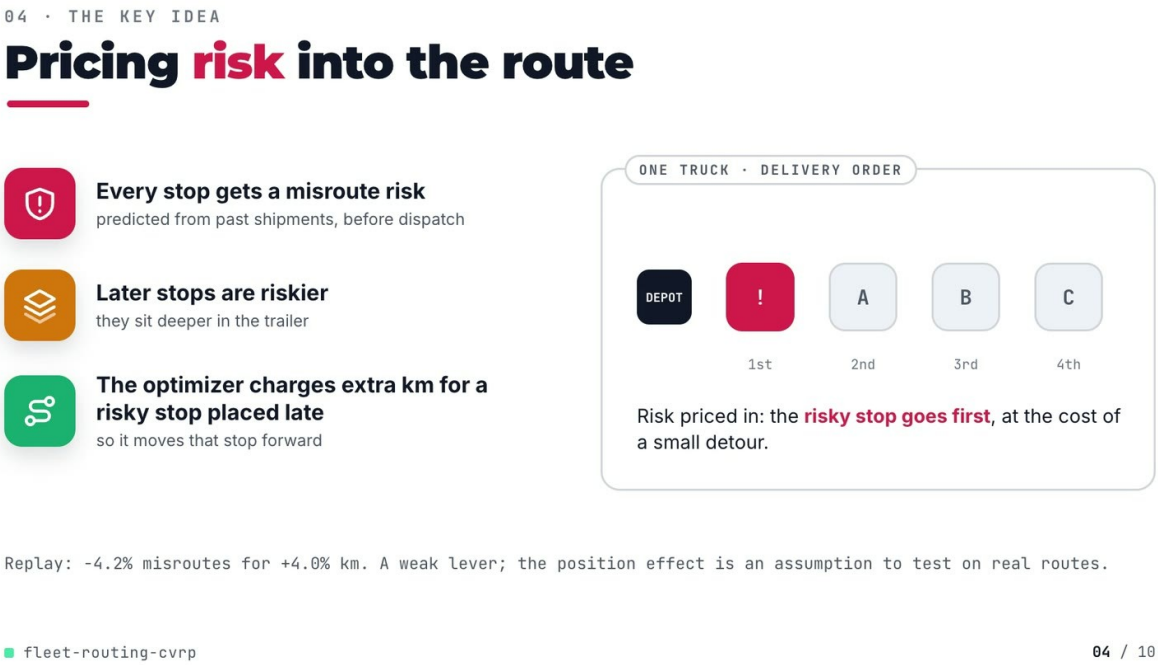
Illustrative, not data:

- Icons and colours are a visual language, not data.
- 'every evening' = each order day is dispatched the next day (A-12).

To change this slide: Edit CONFIG in compositions/explainer.html; badge truth comes from routing_summary.csv (F-11).

Slide 04: The key idea (white chapter, slide B)

19.3-26.3 s · composition `compositions/explainer.html (#ex-b ...)`



On-screen text (verbatim)

04 · THE KEY IDEA

Pricing risk into the route

Every stop gets a misroute risk — predicted from past shipments, before dispatch

Later stops are riskier — they sit deeper in the trailer

The optimizer charges extra km for a risky stop placed late — so it moves that stop forward

ONE TRUCK · DELIVERY ORDER: DEPOT, !, A, B, C (1st..4th). Risk priced in: the risky stop goes first, at the cost of a small detour.

Replay: −4.2% misroutes for +4.0% km. A weak lever; the position effect is an assumption to test on real routes.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-13	Later stops are riskier	`src/fleet_routing/data.py:238-241`; `outputs/full/results.json` `position_effect_logodds_per_stop` · 0.09 generated, 0.094 estimated	estimated slope > 0	✓
F-14	-4.2% misroutes	`outputs/full/routing_summary.csv` · window=peak, misroute_pct: ortools_risk 4.3871, ortools 4.5810	(4.3871 - 4.5810) / 4.5810 x 100 = -4.23%	✓
F-15	+4.0% km	`outputs/full/routing_summary.csv` · window=peak, total_km: ortools_risk 113,937.7, ortools 109,607.3	(113,937.7 - 109,607.3) / 109,607.3 x 100 = +3.95%	✓

Assumptions: A-02, A-03, A-15

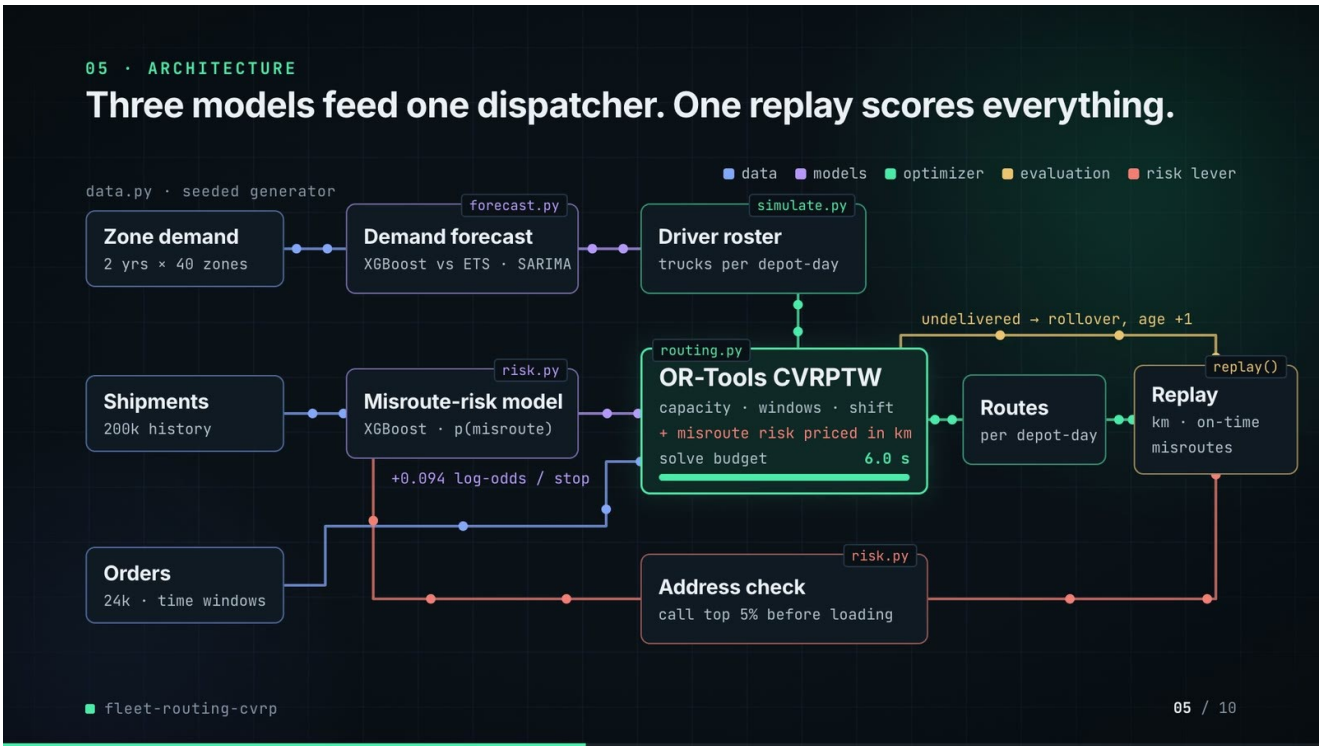
Illustrative, not data:

- The A/B/C/! demo is schematic; it mirrors tests/test_risk_pricing_moves_a_risky_stop_to_the_front, not a real route.

To change this slide: Markup under #ex-b in compositions/explainer.html; the footnote figures are F-14 and F-15.

Slide 05: Architecture

26.0-38.8 s · composition `compositions/s03-architecture.html`



On-screen text (verbatim)

05 · ARCHITECTURE

Three models feed one dispatcher. One replay scores everything.

data.py · seeded generator: Zone demand (2 yrs × 40 zones), Shipments (200k history), Orders (24k · time windows)

forecast.py Demand forecast (XGBoost vs ETS · SARIMA) → simulate.py Driver roster (trucks per depot-day)

risk.py Misroute-risk model (XGBoost · p(misroute)) · +0.094 log-odds / stop

routing.py OR-Tools CVRPTW (capacity · windows · shift · + misroute risk priced in km · solve budget 6.0 s) → Routes (per depot-day) → replay() Replay (km · on-time · misroutes)

undelivered → rollover, age +1 · risk.py Address check (call top 5% before loading)

Legend: data · models · optimizer · evaluation · risk lever

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	300 trucks	`outputs/full/results.json` · trucks	-	✓
F-09	OR-Tools, 6 s per depot / solve budget 6.0 s	`outputs/full/results.json` · solver_seconds_per_depot_day	-	✓
F-16	2 yrs x 40 zones	`data/full/zone_demand.csv`; `outputs/full/results.json` zones · 731 daily rows 2024-01-01..2025-12-31; zones = 40	-	✓
F-17	200k history	`data/full/shipments_history.csv` · row count	-	✓
F-18	24k	`outputs/full/results.json` · dispatch_orders = 24,394	24,394 rounded to thousands	✓
F-19	+0.094 log-odds / stop	`outputs/full/results.json` · position_effect_logodds_per_stop	-	✓
F-20	forecast.py, risk.py, simulate.py, routing.py, replay(), data.py	`src/fleet_routing/` · files + def replay	-	✓
F-21	call top 5% before loading	`src/fleet_routing/config.py:93` · ADDRESS_CHECK_SHARE = 0.05	-	✓

Assumptions: A-01, A-02, A-04, A-09, A-12

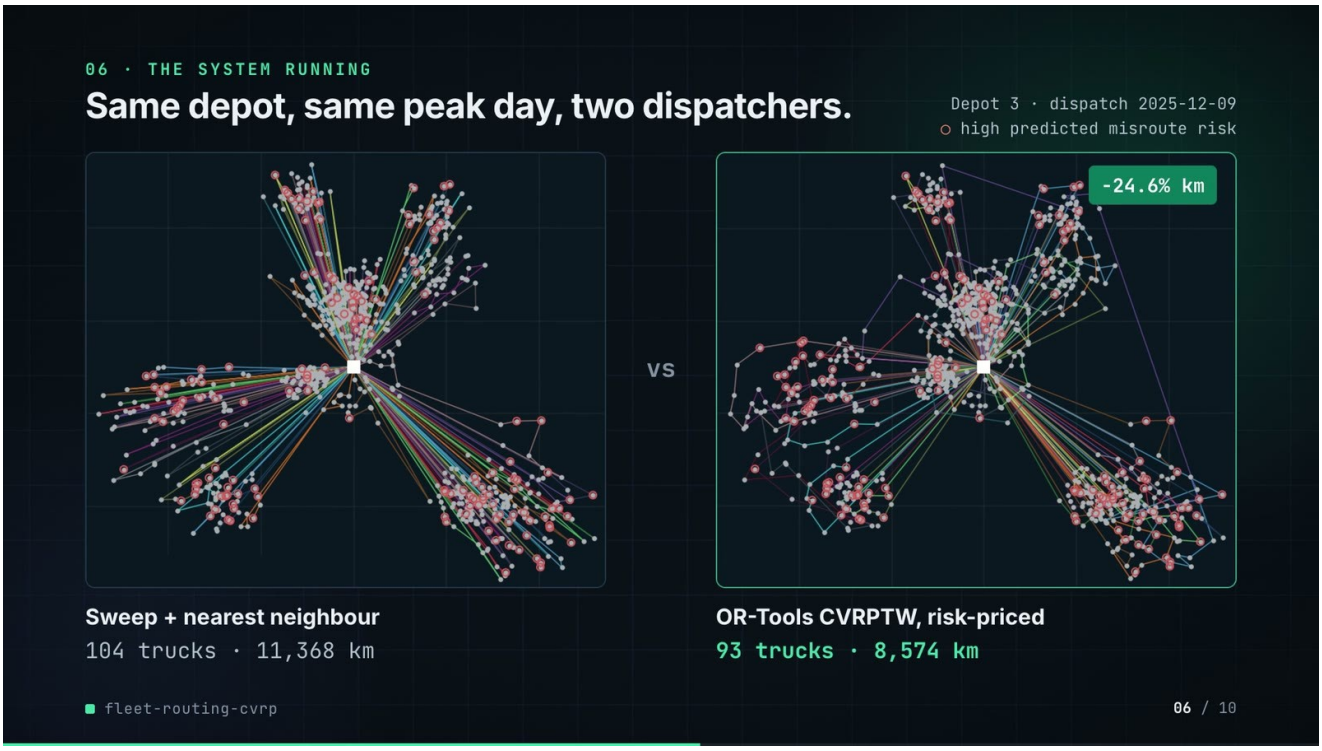
Illustrative, not data:

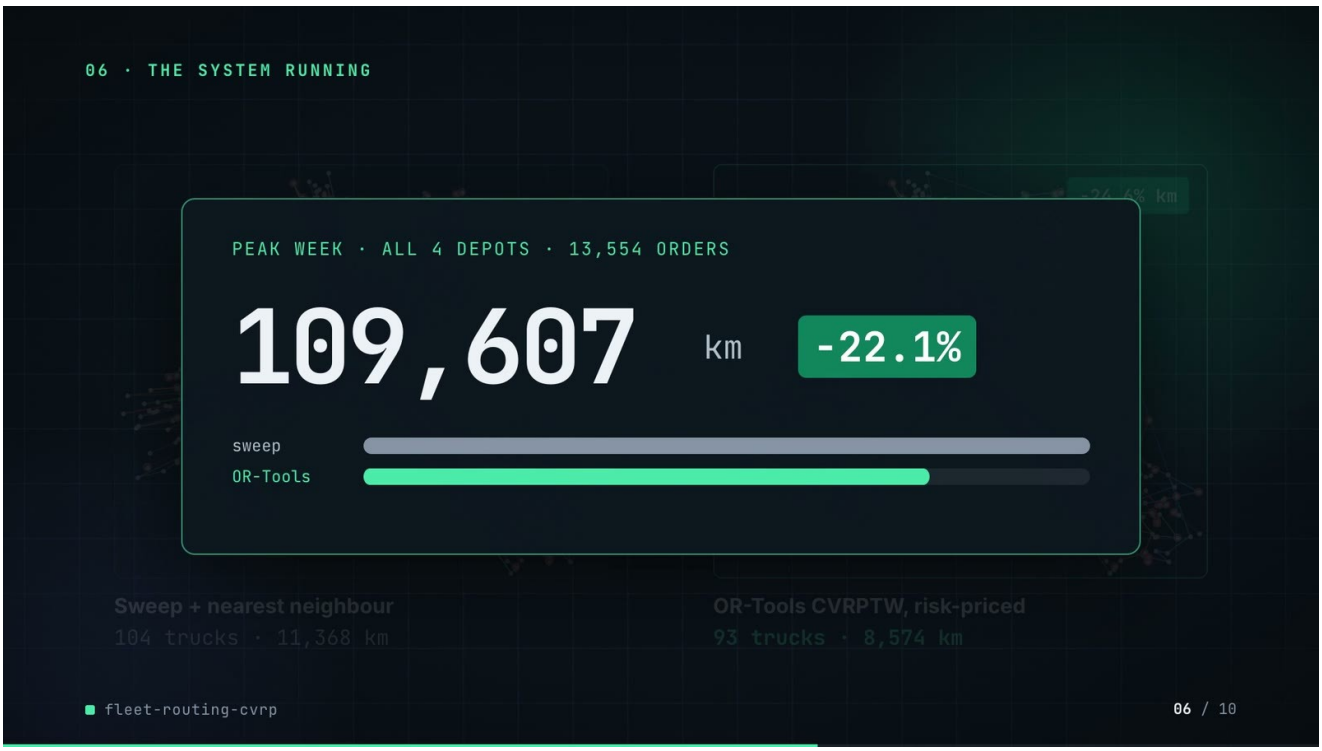
- Packets moving along the wires are decoration; the solve-budget bar fills in real seconds.

To change this slide: Node boxes (#n-*) and the wires array in compositions/s03-architecture.html.

Slide 06: The system running

38.4-53.0 s · composition `compositions/s04-routes.html; assets/map-sweep.png, assets/map-ortools.png (from outputs/full/route_map.png via tools/darkmap.py)`





On-screen text (verbatim)

06 · THE SYSTEM RUNNING

Same depot, same peak day, two dispatchers.

Depot 3 · dispatch 2025-12-09 · ○ high predicted misroute risk

Sweep + nearest neighbour: 104 trucks · 11,368 km | OR-Tools CVRPTW, risk-priced: 93 trucks · 8,574 km | −24.6% km

PEAK WEEK · ALL 4 DEPOTS · 13,554 ORDERS — 140,683 → 109,607 km, −22.1% (sweep vs OR-Tools bars)

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-22	Depot 3 · dispatch 2025-12-	`data/full/depots.csv (busiest depot);`	-	✓

ID	Shown	Source · key	Calculation	Check
	09	`src/fleet_routing/cli.py:121-125; config peak window` · depot with most vehicles = 3; first peak dispatch = 2025-12-08 + 1 day		
F-23	104 trucks	`outputs/full/route_map.png (left panel title)` · plot title	-	manual
F-24	11,368 km	`outputs/full/route_map.png (left panel title)` · plot title	-	manual
F-25	93 trucks	`outputs/full/route_map.png (right panel title); re-run in demo assets/dispatch_sheet.json` · trucks_used	-	✓
F-26	8,574 km	`outputs/full/route_map.png (right panel title)` · plot title	-	manual
F-27	-24.6% km	`F-24, F-26 (plot titles)` · 8,574 vs 11,368	(8,574 - 11,368) / 11,368 x 100 = -24.58%	✓
F-28	13,554 orders	`outputs/full/routing_summary.csv` · arm=sweep, window=peak, orders	-	✓
F-29	140,683 km	`outputs/full/routing_summary.csv` · arm=sweep, window=peak, total_km	-	✓
F-30	109,607 km	`outputs/full/routing_summary.csv` · arm=ortools, window=peak, total_km	-	✓
F-31	-22.1%	`outputs/full/routing_summary.csv` · F-30 vs F-29	(109,607.3 - 140,683.4) / 140,683.4 x 100 = -22.09%	✓

F-27 note: Arithmetic is checked; the two inputs are manual (F-24, F-26).

Assumptions: A-03, A-04, A-05, A-12, A-14

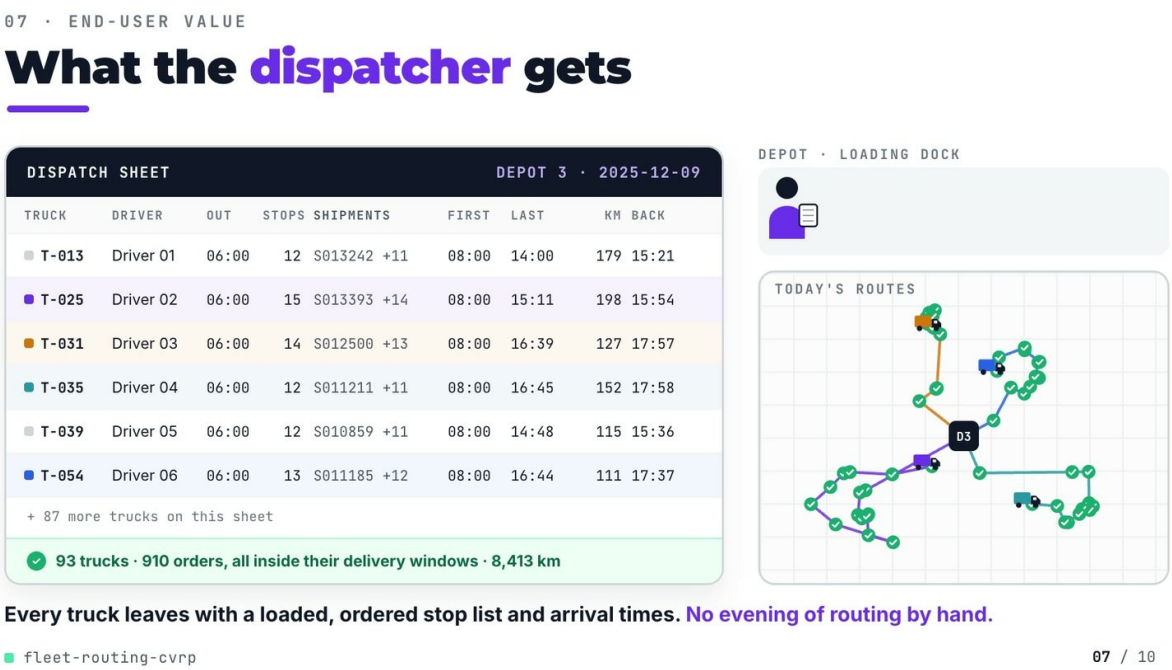
Illustrative, not data:

- Maps are the repo's real route_map.png, colour-inverted to dark; the legend is redrawn because the original legend was removed from the image.
- The map compares sweep with risk-priced OR-Tools; the peak-week counter compares sweep with plain OR-Tools.

To change this slide: Map captions (#routes-cap-l/r, #routes-chip) and counter (#routes-wk-num, #routes-bar) in compositions/s04-routes.html; maps come from make run → outputs/full/route_map.png.

Slide 07: End-user value (white)

52.7-65.7 s · composition `compositions/value.html` (SHEET block injected by `tools/inject_sheet.py`)



On-screen text (verbatim)

07 · END-USER VALUE

What the dispatcher gets

DISPATCH SHEET · DEPOT 3 · 2025-12-09 — columns TRUCK, DRIVER, OUT, STOPS, SHIPMENTS, FIRST, LAST, KM, BACK

T-013 Driver 01 06:00 12 S013242 +11 08:00 14:00 179 15:21

T-025 Driver 02 06:00 15 S013393 +14 08:00 15:11 198 15:54

T-031 Driver 03 06:00 14 S012500 +13 08:00 16:39 127 17:57

T-035 Driver 04 06:00 12 S011211 +11 08:00 16:45 152 17:58

T-039 Driver 05 06:00 12 S010859 +11 08:00 14:48 115 15:36

T-054 Driver 06 06:00 13 S011185 +12 08:00 16:44 111 17:37

+ 87 more trucks on this sheet · ✓ 93 trucks · 910 orders, all inside their delivery windows · 8,413 km

DEPOT · LOADING DOCK · TODAY'S ROUTES

Every truck leaves with a loaded, ordered stop list and arrival times. No evening of routing by hand.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-32	93 trucks	`demo assets/dispatch_sheet.json (re-run of the repo's routing for Depot 3, 2025-12-09)` · trucks_used (rostered 104)	-	✓
F-33	910 orders	`demo assets/dispatch_sheet.json; data/full/dispatch_orders.csv` · orders on 2025-12-08 for depot 3	-	✓
F-34	all inside their delivery windows	`demo assets/dispatch_sheet.json` · on_time = delivered = orders	910 / 910	✓
F-35	8,413 km	`demo assets/dispatch_sheet.json` · total_km = 8,412.7	rounded	✓
F-36	T-013, T-025, T-031, T-035, T-039, T-054	`demo assets/dispatch_sheet.json, injected into compositions/value.html` · vehicles 12, 24, 30, 34, 38, 53	truck id = vehicle index + 1; depart = SHIFT_START	✓
F-37	+ 87 more trucks	F-32 · 93 - 6 rows	93 - 6 = 87	✓

Assumptions: A-03, A-04, A-05, A-06, A-10, A-11

Illustrative, not data:

- Driver 01-06 are placeholders (no real people).
- Dispatcher figure, loading dock, truck drawings, 3 boxes per truck (stands in for 12-15 shipments) and driving speed are illustrative.

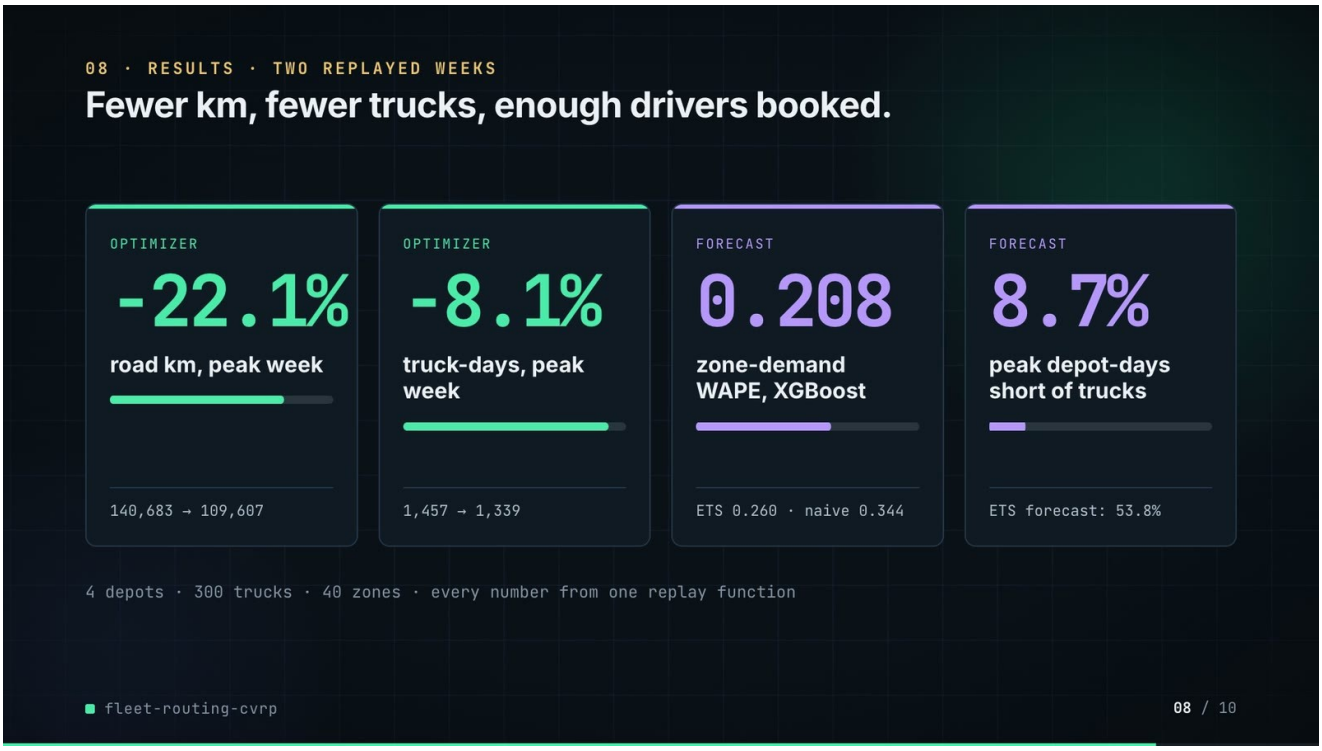
• The 4 animated trucks (T-025, T-031, T-035, T-054) were chosen because their routes spread in different directions; their stop positions and order are real, scaled to fit the panel.

• 'No evening of routing by hand' is framing from the README problem statement, not a measured result.

To change this slide: Re-run tools/export_dispatch_sheet.py, then tools/inject_sheet.py (--rows / --animate pick the trucks); layout in compositions/value.html.

Slide 08: Results

65.4-74.2 s · composition `compositions/s05-results.html (stats array)`



On-screen text (verbatim)

08 · RESULTS · TWO REPLAYED WEEKS

Fewer km, fewer trucks, enough drivers booked.

OPTIMIZER −22.1% road km, peak week · 140,683 → 109,607

OPTIMIZER −8.1% truck-days, peak week · 1,457 → 1,339

FORECAST 0.208 zone-demand WAPE, XGBoost · ETS 0.260 · naive 0.344

FORECAST 8.7% peak depot-days short of trucks · ETS forecast: 53.8%

4 depots · 300 trucks · 40 zones · every number from one replay function

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	300 trucks	`outputs/full/results.json` · trucks	-	✓
F-04	4 depots	`outputs/full/results.json` · depots	-	✓
F-16	2 yrs x 40 zones	`data/full/zone_demand.csv`; `outputs/full/results.json zones` · 731 daily rows 2024-01-01..2025-12-31; zones = 40	-	✓
F-29	140,683 km	`outputs/full/routing_summary.csv` · arm=sweep, window=peak, total_km	-	✓
F-30	109,607 km	`outputs/full/routing_summary.csv` · arm=ortools, window=peak, total_km	-	✓
F-31	-22.1%	`outputs/full/routing_summary.csv` · F-30 vs F-29	(109,607.3 - 140,683.4) / 140,683.4 x 100 = -22.09%	✓
F-38	-8.1%	`outputs/full/routing_summary.csv` · vehicle_days: ortools 1,339 vs sweep 1,457	(1,339 - 1,457) / 1,457 x 100 = -8.10%	✓
F-39	1,457	`outputs/full/routing_summary.csv` · arm=sweep, window=peak, vehicle_days	-	✓
F-40	1,339	`outputs/full/routing_summary.csv` · arm=ortools, window=peak, vehicle_days	-	✓
F-41	0.208	`outputs/full/forecast_scores.csv` · model=xgboost, period=all, wape	-	✓
F-42	ETS 0.260	`outputs/full/forecast_scores.csv` · model=ets, period=all, wape	-	✓
F-43	naive 0.344	`outputs/full/forecast_scores.csv` · model=seasonal_naive, period=all, wape	-	✓
F-44	8.7%	`outputs/full/roster_scores.csv` · model=xgboost, period=peak, short_depot_days_pct	-	✓
F-45	ETS forecast: 53.8%	`outputs/full/roster_scores.csv` · model=ets, period=peak, short_depot_days_pct	-	✓

Assumptions: A-04, A-05, A-10, A-12, A-13, A-15

Illustrative, not data:

- Mini bars under each number are drawn to the same ratio (after / before).

To change this slide: The stats array and card text in compositions/s05-results.html.

Slide 09: What didn't win

73.8-80.0 s · composition `compositions/s06-twist.html`



On-screen text (verbatim)

09 · WHAT DIDN'T WIN

The XGBoost risk model ties a two-line supervisor rule.

Peak misroutes cut by calling the top 5% of consignees before loading

Supervisor rule (postcode, then address) 16.6% · XGBoost (ROC-AUC 0.645) 16.4% · Logistic regression (ROC-AUC 0.647) 15.8% · Random 5% 1.1%

Why: the label caps both models (ceiling AUC 0.647). Next step: train on fixable faults only.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-21	call top 5% before loading	`src/fleet_routing/config.py:93` · ADDRESS_CHECK_SHARE = 0.05	-	✓
F-46	16.6%	`outputs/full/address_checks.csv` · ranked_by=rule, window=peak, reduction_pct	-	✓
F-47	16.4%	`outputs/full/address_checks.csv` · ranked_by=xgboost, window=peak, reduction_pct	-	✓
F-48	15.8%	`outputs/full/address_checks.csv` · ranked_by=logistic, window=peak, reduction_pct	-	✓
F-49	1.1%	`outputs/full/address_checks.csv` · ranked_by=random, window=peak, reduction_pct	-	✓
F-50	ROC-AUC 0.645	`outputs/full/risk_scores.csv` · model=xgboost, period=all, roc_auc	-	✓
F-51	ROC-AUC 0.647	`outputs/full/risk_scores.csv` · model=logistic, period=all, roc_auc	-	✓
F-52	ceiling AUC 0.647	`outputs/full/risk_scores.csv` · model=ceiling, period=all, roc_auc	-	✓

Assumptions: A-02, A-08, A-09, A-15

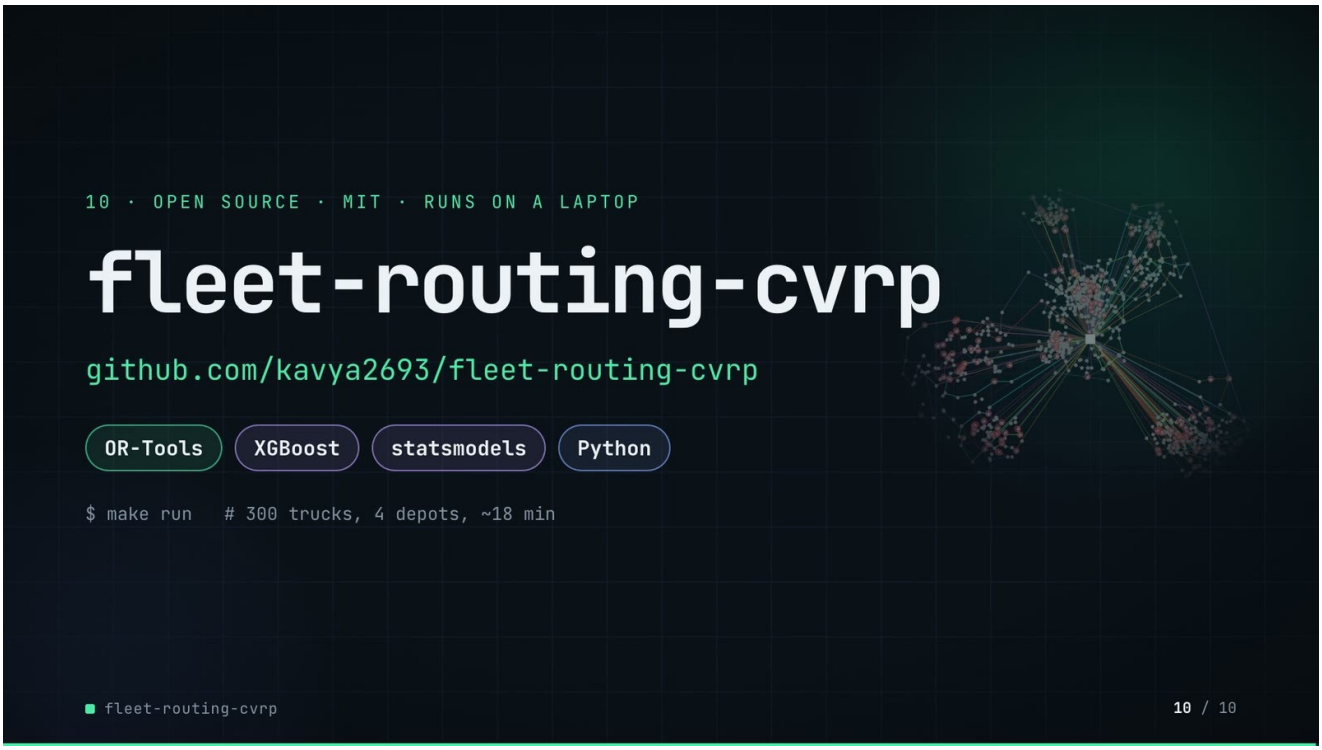
Illustrative, not data:

- Bar lengths: 1,000 px track = 20% reduction.

To change this slide: Rows #twist-r1..r4 (bar widths are inline styles) in compositions/s06-twist.html.

Slide 10: End card

79.6-84.0 s · composition `compositions/s07-end.html`



On-screen text (verbatim)

10 · OPEN SOURCE · MIT · RUNS ON A LAPTOP
fleet-routing-cvrp
github.com/kavya2693/fleet-routing-cvrp
OR-Tools · XGBoost · statsmodels · Python
$ make run # 300 trucks, 4 depots, ~18 min

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-53	MIT	`LICENSE` line 1 · -	-	✓
F-54	~18 min	`outputs/full/results.json` · runtime_seconds = 1063	1063 / 60 = 17.7 min	✓
F-55	OR-Tools · XGBoost · statsmodels · Python	`pyproject.toml` dependencies · ortools, xgboost, statsmodels	-	✓
F-56	$ make run	`Makefile` · run target	-	✓

Assumptions: A-15

Illustrative, not data:

- Faded route map in the background is the OR-Tools panel of route_map.png.

To change this slide: Text in compositions/s07-end.html.

5. Change-impact matrix

Input / assumption	Affected figures	Slides	Files to edit	Re-run
A-01: All data is synthetic, produced by a seeded generator sized to the production problem (300 trucks, 4 depots, 40 zones, 2 years of demand, 200,000 historical shipments)	all figures	01, 02, 03, 04, 05, 06, 07, 08, 09, 10	`src/fleet_routing/config.py (FULL profile); src/fleet_routing/data.py`	make data; make run
A-02: Misroute risk rises with route position: the generator adds 0	F-13, F-14, F-15, F-19, F-46, F-47, F-48, F-49	04, 05, 09	`src/fleet_routing/data.py:241`	make data; make run
A-03: A misroute is priced at 150 km inside the risk-priced router (cost per stop = 150 km x 0	F-14, F-15, F-23, F-24, F-25, F-26, F-27, F-32, F-33, F-34, F-35, F-36	04, 06, 07	`src/fleet_routing/config.py:90`	make run; export_dispatch_sheet.py + inject_sheet.py
A-04: Each OR-Tools solve is capped at 6 s wall-clock per depot-day (1	F-09, F-25, F-26, F-30, F-31, F-35, F-38, F-40	03, 05, 06, 07, 08	`src/fleet_routing/config.py:45`	make run; export_dispatch_sheet.py + inject_sheet.py
A-05: Distances are straight lines x 1	F-24, F-26, F-27, F-29, F-30, F-31, F-35, F-36	06, 07, 08	`src/fleet_routing/ config.py:74-76`	make run; export_dispatch_sheet.py + inject_sheet.py
A-06: Drivers leave the depot from 06:00 and must be back by 18:00	F-07, F-36	02, 07	`src/fleet_routing/ config.py:72-73`	make run; export_dispatch_sheet.py + inject_sheet.py
A-07: Delivery windows are hard constraints in OR-Tools (it drops a stop rather than arrive late) and are ignored by the sweep baseline, so the on-time comparison is lopsided by design	F-11, F-34	03, 07	`src/fleet_routing/routing.py`	make run
A-08: A pre-dispatch call fixes the postcode, address and handwritten-label faults completely; it cannot fix a between-depots zone or a cross-dock relabel	F-46, F-47, F-48, F-49	09	`src/fleet_routing/risk.py:134-142`	make run
A-09: The dispatch office calls the top 5% of each day's consignees	F-21, F-46, F-47, F-48, F-49	05, 09	`src/fleet_routing/config.py:93`	make run
A-10: Trucks booked per depot-day = ceil(forecast orders x 2	F-44, F-45, F-32	07, 08	`src/fleet_routing/ config.py:82-84`	make run
A-11: Fleet mix: 70% tractors with 26 pallets, 30% rigids with 14 pallets; depot fleets 63 / 86 / 47 / 104	F-03, F-23, F-25, F-32, F-36	02, 05, 06, 07, 08	`src/fleet_routing/ config.py:67-69`	make data; make run
A-12: Evaluation windows: quiet week 2025-10-13	F-28, F-28, F-29, F-30, F-31, F-38, F-39, F-40	06, 08	`src/fleet_routing/config.py:43`	make run
A-13: Forecasts are backtested from 12 weekly (Monday) origins over the last 12 weeks of 2025; every feature is at least a week old	F-10, F-41, F-42, F-43, F-44, F-45	03, 08	`src/fleet_routing/config.py:44`	make run
A-14: Router cost terms: a dropped stop costs 1,000 km x (1 + days waited); each truck used costs 40 km	F-25, F-26, F-30, F-35, F-40	06, 07, 08	`src/fleet_routing/ routing.py:32-34`	make run
A-15: Video rounding: percentages to 1 decimal, km and counts to whole numbers with thousands separators, WAPE and AUC to 3 decimals; '24k', '200k', '~18 min' are rounded labels	F-14, F-15, F-18, F-27, F-31, F-38, F-41, F-42, F-43, F-44, F-45, F-46, F-47, F-48, F-49, F-50, F-51, F-52, F-54	04, 05, 06, 08, 09, 10	`the composition file named on each slide`	hyperframes render
A-16: Slide 01 (industry, everyday problems) is general logistics knowledge used as context	F-01, F-02	01	`compositions/s01-context.html`	hyperframes render

6. Known differences and caveats

• Map-day km: the slide shows 8,574 km (route_map.png title, from make run). The single-day re-run used for slide 07 gives 8,412.7 km. OR-Tools stops on wall-clock time, so repeated runs differ slightly (README 'Known limitations': 87,715 vs 88,698 km on the quiet week).

• −24.6% (slide 06) is derived from the two plot titles; the repo does not store per-depot-day routes for make run.

• The map (slide 06) compares sweep with risk-priced OR-Tools; the peak-week counter (slides 06, 08) compares sweep with plain OR-Tools.

• 'Up to 2,800 orders a day' (slide 02) is the busiest day in the data (2,785 orders, rounded up); the README's 'closer to 2,700' describes a typical peak day.

• On-time 100% / 'windows kept' is true by construction: OR-Tools treats windows as hard and rolls stops over instead (A-07).

- Slide 01 statements are general industry knowledge (A-16).
- Route-position misroute risk is a mechanism written into the generator (A-02); its real size must be measured on real route histories before the routing-side misroute numbers mean anything (README 'Known limitations').
- Driver names on slide 07 are placeholders.

Appendix: environment

Item	Value
Commit	`c9b3c09d094cd863c61c2af426513b6d9b33ba68`
Python	3.11.7
ortools	9.15.6755
xgboost	3.2.0
pandas	3.0.6
numpy	2.4.6
statsmodels	0.15.0
sklearn	1.9.1
Profile / seeds	profile=full, seed=7, risk_model_seed=7
Run time	results.json runtime_seconds = 1063 s (about 17.7 min) for make run on the author's laptop